

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electrical Engineering) SEMESTER - 6 Summer 2026 (Regular)

Course :Bachelor of Technology (Electrical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BTEEC603 - CONTROL SYSTEM ENGINEERING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

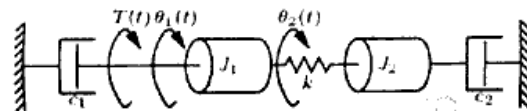
- 1 In an open loop control system
 - (a) Output is independent of control input
 - ☒ (b) Output is dependent on control input
 - (c) Only system parameters have effect on the control output
 - (d) None of the above
- 2 A control system in which the control action is somehow dependent on the output is known as
 - ☒ (a) Closed loop system
 - (b) Semiclosed loop system
 - (c) Open system
 - (d) None of the above
- 3 Which of the following is an open loop control system ?
 - ☒ (a) Field controlled D.C. motor
 - (b) Ward leonard control
 - (c) Metadyne
 - (d) Stroboscope
- 4 has tendency to oscillate.
 - (a) Open loop system
 - ☒ (b) Closed loop system
 - (c) Both (a) and (b)
 - (d) Neither (a) nor (b)
- 5 A good control system has all the following features except
 - (a) good stability
 - ☒ (b) slow response
 - (c) good accuracy
 - (d) sufficient power handling capacity
- 6 Transfer function of a system is used to calculate which of the following ?
 - (a) The order of the system
 - (b) The time constant
 - ☒ (c) The output for any given input
 - (d) The steady state gain
- 7 The transient response, with feedback system,
 - (a) rises slowly
 - (b) rises quickly
 - (c) decays slowly
 - ☒ (d) decays quickly

- 8 Which of the following statements is correct for any closed loop system ?
 (a) All the co-efficients can have zero value
 (B) All the co-efficients are always non-zero
~~(c)~~ Only one of the static error co-efficients has a finite non-zero value
 (d) None of the above
- 9 In force-voltage analogy, velocity is analogous to
~~(a)~~ current (b) charge (c) inductance (d) capacitance
- 10 Technique gives quick transient and stability response
 (a) Root locus (b) Bode (c) Nyquist (d) Nichols
- 11 A control system is generally met with the time response specifications:
 a) Damping factor b) Setting time c) Steady state accuracy d)
 All of the mentioned
- 12 In closed loop control system, with positive value of feedback gain the overall gain of the system will
~~(a)~~ decrease (b) increase (c) be unaffected (d) any of the above

Q2. Solve the following.

- A) Determine transfer function of given system.

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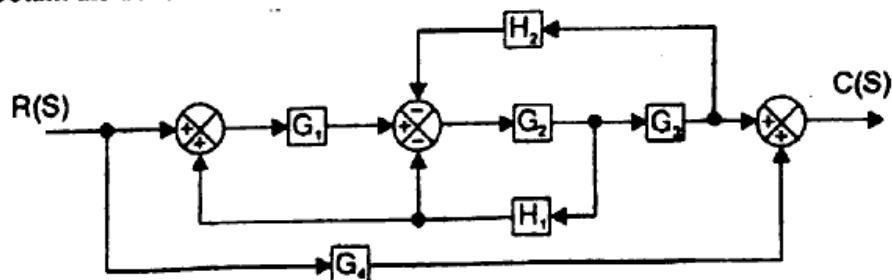
- B) Compare open loop system with close loop control system.

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Q3. Solve the following.

- A) Obtain the transfer function

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- B) Find the steady state error for unit step, unit ramp and unit acceleration inputs for the following systems.

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$$\frac{1000}{s^2(s+1)(s+20)}$$

Q4. Solve Any Two of the following.

- A) Using Routh's stability criterion determine stability of given system.

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$$S^6 + 3S^5 + 5S^4 + 9S^3 + 8S^2 + 6S + 4 = 0$$

- B) For the system having closed loop transfer function

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$$\frac{C(s)}{R(s)} = \frac{18}{s^2 + 4s + 18}$$

- Determine i) Damped frequency of oscillations
ii) Peak time
iii) % peak overshoot

- C) Plot the root locus of unity feedback control system has open loop transfer function

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$$G(s) = \frac{K}{s(s+4)}$$

Q5. Solve Any Two of the following.

- A) Explain PI Controller and PD Controller with its advantages and disadvantages. 6
B) Determine gain margin and phase margin using bode plot and comment on stability. 6

$$G(s) = \frac{10}{s(1 + 0.5s)(1 + 0.01s)}$$

- C) Write short note on PID controller

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Q6. Solve Any Two of the following.

- A) Obtain the transfer function of the system defined by

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$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

- B) Find state space model equation for given system

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$$\frac{Y(s)}{U(s)} = \frac{1}{s^3 + 4s^2 + 3s + 3}$$

- C) Verify controllability and observability of control system which is represented in state space model. 6

$$A = \begin{bmatrix} 0 & -2 \\ 1 & -3 \end{bmatrix} \quad B = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad C = [0 \quad 1]$$

*** End ***